

Super-Efficient Electric Ceiling Fans with Infrared Remote Controls

Versa Drives has come up with super-efficient electric ceiling fans, which consume less than half of the energy of similar-sized ordinary fans. Moreover, they are equipped with infrared remote controls and are available for consumers in multiple variants

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In a tropical country like India where energy-hungry common appliances are used continuously round the year, super-efficient ceiling fans can have an enormous effect on energy savings for consumers. So presented here is a fan that consumes less than half the power of an ordinary fan and thus proves to be an investment that saves huge amount of energy and money in its lifetime. It has been designed and manufactured by Versa Drives Pvt Ltd and is currently being sold under the brand name 'Superfan.'

Superfan also comes with a remote control powered by infrared technology, which not only provides ease of use but also enhances efficiency by eliminating regulator and its associated power losses.

When asked what prompted them to design an energy-efficient ceiling fan, Sundar Murugandhan, managing director, Versa Drives, says, "Most people are not aware that ceiling fans in India consume 12,000MW of electricity. With an efficient ceiling fan like Superfan, consumers can save thousands of megawatts of power for India by reducing this demand to 6000MW."

Challenges

The conventional ceiling-type fans available in the market use a single-phase induction motor with a capacitor to work from standard 230V supply—these are inexpensive but



Superfan with remote

at the same time very inefficient as well. The main challenge was to run a fan with a motor that will consume less energy whilst remaining cost-effective. The efficient permanent magnet brushless DC (BLDC) motor was one of the options but it was expensive and posed many cost challenges. This motor had the additional requirement of sensors to indicate rotor positions to the electronic drive. Also, an electronic drive comes with the disadvantage of introducing harmonics to the power source which, if not controlled, would reduce the effectiveness of a power-efficient system, especially if used in large numbers as in the case of ceiling fans.

Talking about the major cost challenges, Sundar says, "This new design involved a permanent magnet and

an electronic drive which pushed the product costs up. The ceiling fan market in India is so competitive that it has driven the costs down. So building Superfan at reasonably higher costs was a huge challenge. It also had to have the same wiring and installation procedures so that replacement market can be targeted, and electricians need not be specially trained for our product. We had to invest in a lot of tools and find good suppliers to reduce the capital costs."

Efficient design

Superfan has an efficient BLDC motor with ceramic magnets and a rotor back-iron formed by mild steel arc segments. An electronic controller consisting of a switched-mode power supply, an inverter and a micro-